

Multimodal Car Driver Stress Recognition

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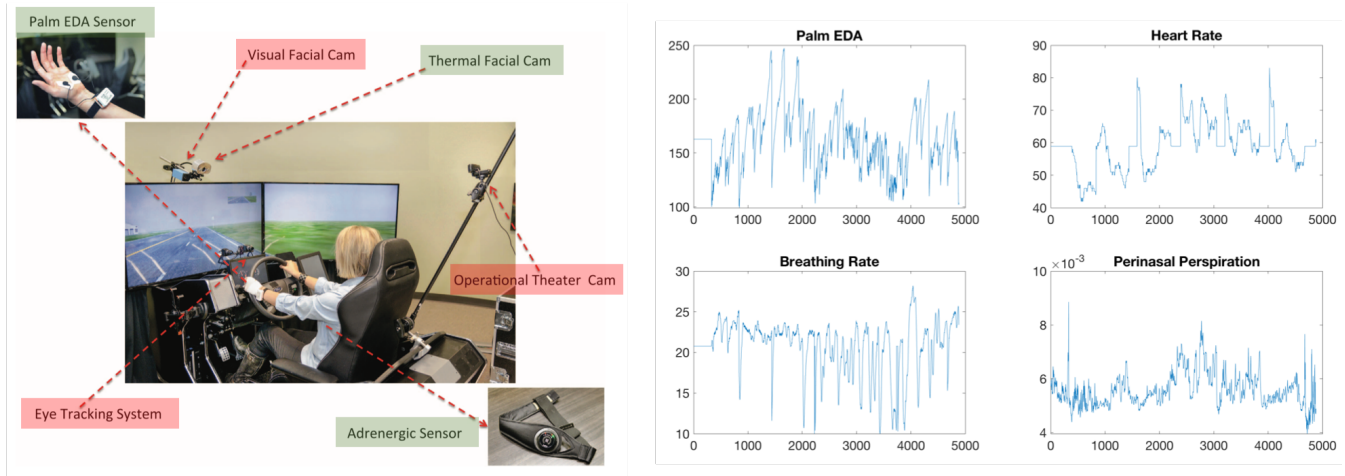


Figure 1: Left side of the figure: depiction of the experimental setup adopted in [20]; highlighted in green the sensors used to acquire the physiological data used in this paper. Right side of the figure: graphical representation of the physiological signals acquired during a driving session of the experiments presented in [20].

ABSTRACT

In this paper we address the problem of multimodal car driver stress recognition. To this aim, four different signals are considered: heart rate (HR), breathing rate (BR), palm EDA (P-EDA), and perinasal perspiration (PER-EDA). The raw signals are windowed and for each window 21 different features, including both time-domain and frequency-domain descriptors, are extracted. The recognition problem is formulated as a stress vs no-stress binary problem, and is addressed in two different experimental setups: five-fold cross validation and leave one subject out. In both setups the extracted features are classified, both individually and concatenated, with three different classifiers (k -NN, SVM, and ANN) using them both alone and stacking their predictions. Experiments run on a publicly available database of multimodal signals acquired in a controlled experiment on a driving simulator show that the best recognition

results are obtained feeding the classifiers with the concatenation of the features of all the signals considered, reaching a micro average accuracy of 77.25% and 65.09% in the two experimental setups respectively.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; *Supervised learning by classification*.

KEYWORDS

Driver fatigue, stress detection, machine learning

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1 INTRODUCTION AND BACKGROUND

Based on a report published in 2012 by the National Highway Traffic Safety Administration office of the U.S. Department of Transportation, the most common causes of car accidents are: drivers' inattention, high-speed driving, drunken driving, misperception,

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decision errors, and driver incapacitation (e.g., falling asleep or having a heart attack while driving) [8]. Even more, more recent studies report that in US an average of 328,000 traffic accidents annually involve a drowsy driver, while in Europe, 20-25% of total traffic accidents were due to drowsy drivers [6, 21]. In this context, driver status monitoring (DSM) systems have recently emerged as possible way to prevent car accidents [3]. In particular, the design and evaluation of automatic DSM systems based on heterogeneous sensors, such as physiological sensors, video cameras and car telemetries, are among the most explored topics in the field of car and road safety also known as advanced driver-assistance systems (ADAS) [10].

These topics have recently attracted a great attention thanks to the increase of several consumer-ready products for the measurement of physiological parameters as well as thanks to recent technological developments in the field of Internet of Things (IoT) and Embedded Systems (ES). DSM systems are mostly based on machine learning and thus by definition they need data to be developed and validated, especially now with the recent success of the deep learning. A recent paper published a large amount of physiological data and telemetries acquired by people engaged in simulated driving tasks with different levels of stress, with the aim to enable research into driving behaviors under neatly abstracted distracting stressors [20].

High level of stress compromises decision-making skills of a driver, thus his performance and awareness drastically decrease [22]. For example, the physiological response of the human body to the stress causes high levels of activation of the sympathetic nervous system and consequently causes: dilatation of the pupil, release of endorphins, increased cardiac and respiratory rates, adrenaline secretion, dilation of the arterioles and constriction of the veins [1, 5]. This inevitably affects car and road safety.

Several automatic stress monitoring systems that exploit human physiological parameters have been proposed in literature. The signals considered include heart-rate (HR), heart-rate variability (HRV), pupil-dilation, blood-pressure, respiration rate (BR) and Galvanic Skin Response (GSR) [4, 7, 9, 11, 13, 13–15, 17, 18, 23, 24]. Several studies showed a strong correlation between ECG, EDA, BR and driver stress levels [17]. In particular, the paper by Ollander *et al.* [16] investigated the use of a Naïve Bayes classifier and a combination of features extracted from HR, BR and GSR to predict stress level. The experiments have been conducted exploiting the database of real driving sessions proposed by Picard *et al.* [7] and the best accuracy reached is about 87%. Napoletano *et al.* [15] combined HR and BR of the the database published by Taamneeh *et al.* [20] to classify stress vs no-stress using Support Vector Machines and K-Nearest Neighbour. They obtained an accuracy of about 70%.

In this paper we investigate the use of four different physiological signals to address the problem of multimodal car driver stress recognition. The signals considered are: heart rate (HR), breathing rate (BR), palm EDA (P-EDA), and perinasal perspiration (PER-EDA). Each raw signal is windowed and for each window a set of 21 different features is extracted with both time-domain and frequency-domain descriptors. The extracted features are classified with four different classifiers: k -NN, SVM, ANN and their stacking. Classifiers are fed with both descriptors coming from a single physiological signal and with the concatenation of descriptors coming

from all the four signals considered. The car driver stress recognition task is formulated as a stress vs no-stress binary problem, and is addressed in two different experimental settings: the first one is a stratified five-fold cross validation setup, the second one is the more challenging leave one subject out setup. Experiments are run on a publicly available database of multimodal signals acquired in a controlled experiment on a driving simulator.

2 MATERIALS AND METHODS

2.1 Description of the Database

The database used in this paper has been released in 2017 by Taamneeh *et al.* [20]. It contains multimodal signals acquired in a controlled experiment on a driving simulator that involves $n = 68$ volunteers that drove the same highway under four different conditions: no distraction, cognitive distraction, emotional distraction, and sensorimotor distraction. The database includes also a special driving session, where all subjects experienced a startle stimulus in the form of unintended acceleration-half of them under a mixed distraction, and the other half in the absence of a distraction. Fig. 1 shows the driving simulator set up and some samples extracted from physiological signals recorded during the driving session.

The signals included in the database are speed, acceleration, brake force, steering, and lane position signals, perinasal perspiration (PER-EDA), palm EDA (P-EDA), heart rate (HR), breathing rate (BR), and facial expression signals. Moreover, biographical and psychometric covariates as well as eye tracking data were also obtained. Perinasal perspiratory signals commensurate with electrodermal (EDA) in the palm, for this reason is called perinasal EDA (PER-EDA) [19].

The database includes two sessions of experiments:

- EXPERIMENT-I: focused on the effect of cognitive, emotional and sensorimotor stressors on driving behaviors under typical conditions.
- EXPERIMENT-II: focused on the effect of stress on reactivity to a startling event while driving; this startling event was unintended acceleration.

In this paper we focus on the EXPERIMENT-I and we consider four signals: PER-EDA, P-EDA, HR and BR. Perinasal perspiratory signals have been extracted from the thermal facial videos using the S-Interface software [20] applied to a region of interest. Palm EDA has been acquired with the Shimmer3 GSR sensor (Shimmer, Dublin, Ireland) that has a measurement range of 10-4700 k Ω . HR and BR signals have been acquired with the adrenergic sensor Zephyr BioHarness 3.0 (Zephyr Technology, Annapolis, MD). The sensor detects a heart rate range of 25-240 bpm and a breathing rate range of 4-70 bpm. The adrenergic sensor was connected to a chest strap that was worn underneath the subject's clothing. Fig. 1 shows the PER-EDA, P-EDA, HR and BR signal related to the Experiment-I and Experiment-II of the subject 1.

2.1.1 EXPERIMENT-I. This experiment is composed of 7 subsequent sessions:

- Baseline (B): subjects sat quietly in a dimly lit room, listening to soothing music;
- Practice drive (PD): the subjects familiarized themselves with the driving simulator;

- Relaxing drive (RD): the subjects had to drive following some instructions;
- Loaded Drive (LD): Driving with no secondary activity (no additional stressor);
- Cognitive Drive (CD): Driving under a cognitive stressor. The cognitive stressor was mathematical questions;
- Emotional drive (ED): Driving under an emotional stressor. The emotional stressor was emotionally stirring questions posed orally by the experimenter in two phases. There were two sets of questions: a set with less pointed questions and a set with more pointed questions
- Sensorimotor drive (MD): Driving under a sensorimotor stressor. The sensorimotor stressor was texting back words, sent one by one to the subject's smartphone; this texting exchange took place in two phases.

The last three phases were developed alternating phases with distraction and phases without distractions. We decide to discard the session B because the physiological signals related to this session are not available.

We select the sessions CD, ED and MD as representative of stress conditions and PD, RD and LD as representative of no-stress conditions. This turns the problem of stress recognition in a binary classification problem: stress vs no-stress. Session PD and RD have a duration of about 500 seconds while LD, CD, ED and MD have a duration of about 800 seconds.

2.2 Data cleaning, normalization and segmentation

A cleaning procedure has been applied to the database with the aim of removing outliers signals from the database and removing those subjects with missing or not valid signal values. We removed from the database all the signals that had a peak of the HR, BR and P-EDA signals outside the intervals [40-120], [4-40] and [28-628] respectively, for at least 30% of the signal duration. In case of an amount of invalid values lower than 30%, such invalid values have been substituted by the mean of the signal.

To the scope of this paper, we have explored the use of HR, BR, PER-EDA and P-EDA and their combination to classify stress vs no-stress. Since, multimodal classification is possible only if, for each time instant, a valid value of each one of the four aforementioned physiological signals exists, we discard all the subjects with missing or not valid signals in any of the four signals. The initial number of subjects available is 68, after the cleaning and discarding procedures, the number of subjects is 37.

Before being processed, each signal is segmented by using a sliding window of size W and overlap L . An overlap of 50% means that two subsequent windows have 50% of their values in common. After having investigated several values for W ranging from 50 to 400 seconds and three values of L ranging from 25 to 75 % we decide to set $W = 60$ seconds and $L = 50\%$.

2.3 Proposed classification pipeline

The proposed method includes two main modules: feature extraction and classification.

2.3.1 Feature extraction module. Whatever is the signal, this module extracts the following 21 features from each segment of length W :

- (1) Arithmetic mean of the intensity values;
- (2) Median of the intensity values;
- (3) The fourth moment;
- (4) The fifth moment;
- (5) Standard deviation of the intensity values;
- (6) Variance of the intensity values;
- (7) The fourth standard moment, also known as Kurtosis, of the intensity values;
- (8) Skewness of the intensity values;
- (9) Sum of the intensity values;
- (10) Range between maximum and minimum of the intensity values;
- (11) Maximum of the intensity values;
- (12) Minimum of the intensity values;
- (13) Root mean square error of the intensity values;
- (14) Entropy;
- (15) Interquartile range of the intensity values;
- (16) Spectral power density;
- (17) Mean of the spectral power density;
- (18) Median of the spectral power density;
- (19) Root mean square of the differences between two successive peaks;
- (20) Standard deviation of the intervals between two successive peaks;
- (21) The number of pairs of successive peaks intervals that differ by more than 50 ms.

All the data are divided in training and test before being classified and the procedure for doing that is explained in Section 3. The training features are normalized in a way that each feature component ranges between 0 and 1. The test features are then normalized using the values adopted to normalize the training features.

2.3.2 Classification and combination module. The feature vectors associated to the segments are classified through three different classifiers: k -NN, Support Vector Machines (SVM), and Artificial Neural Networks (ANN). All the classifiers are trained in two different setups, i.e. both on features extracted from the individual signals and on the concatenation of these features.

The k -NN uses Euclidean distance and $k = 1$, thus resulting in the 1-nearest neighbor classifier.

The SVMs used have a Radial Basis Function (RBF) kernel. The optimal values of C and γ are found by grid search.

Due to the small quantity of data available the ANNs used have a shallow architecture. The hidden layers are unsupervisedly trained as Autoencoders, stacked together and are followed by a softmax layer. The number of hidden layers and the number of neurons in each of them, i.e. H_1 and H_2 , are selected by grid search in the range [0, 5, 10, 15, 20, 30, 40, 50] for individual features and [0, 5, 10, 15, 20, 40, 60, 80, 100, 120, 140, 160] for concatenated features. The maximum number of hidden layers is fixed to two. Given the possible ranges for H_1 and H_2 , if $H_1 = 0$ or $H_2 = 0$ the resulting ANN has just one hidden layer. If $H_1 = H_2 = 0$, the resulting ANN has no hidden layers and the input features are directly fed to the softmax layer.

As an additional classifier we consider the stacked classifier, that performs the classification on the basis of the predictions of the k -NN, SVM and KNN through majority vote.

3 RESULTS AND DISCUSSION

The task of multimodal car driver stress recognition is faced in two different experimental setup. The first one adopts a five fold cross validation, i.e. the test set contains unseen data from the same subjects seen during training. The second one adopts a leave-one-subject-out cross validation, i.e. the training set contains the data of all subjects except the data from the test subject thus resulting in a 37-fold cross validation. The experimental results in term of recognition rate (and its standard deviation) are reported respectively in Table 1 and 2 for the 5-fold and one-subject-out cross validation. Since the two classes considered (i.e. stress and no-stress) are imbalanced, the recognition rate is computed as micro accuracy. From the results reported in the tables it is possible to see how the best performance obtained in the first experimental setup (i.e. 77.25%, obtained with all the features concatenated) is about 12% higher than the best in the second experimental setup (i.e. 65.06%, obtained with all the features concatenated), highlighting the difficulty of adapting the recognition model to unseen subjects. This difficulty can be also evinced from the accuracy standard deviations, that in the second setup are on average about 6 points higher than those in the first setup.

Concerning the individual features, we can see that those extracted from the BR and PER-EDA signals are able to better distinguish between stress and no-stress classes. This can be also seen in Figure 2.a, where the box plots of the classification results obtained on all signals in both experimental setups together are reported. Interestingly, considering the best performance on individual signals, the performance drop between the first experimental setup (63.90% obtained by ANN on PER-EDA) and the second experimental setup is only about 1% (62.66% obtained by ANN on BR).

Concerning the classifiers considered, in Figure 2.b we report the boxplot of the performance obtained in the two experimental setups together, grouped by classifier. From the plot it is possible to notice that on average, the best classifier is ANN.

Table 3 shows the accuracy of each class (stress and no-stress) achieved by the best solution, that is the stacking of classifiers with the combination of all physiological signals. It is more accurate to recognize stress than no-stress for both folding strategies. This suggests that the method is more capable to catch anomalies with respect to normal behavior. Further investigation should be done to understand how much the accuracy of the classifier can be improved by turning the binary classification problem in a one class problem, that is stress detected as anomalies [12]. In the case of one subject out cross validation there is a high spread between subjects. It is probably due to the fact that P-EDA values range in a very large interval, thus subjectivity of the signal may affect the recognition rate.

4 CONCLUSION

In this paper we investigated the use of heart rate (HR), breathing rate (BR), palm EDA (P-EDA), and perinasal perspiration (PER-EDA) for the automatic multimodal recognition of car driver stress.

The raw signals are windowed and for each window 21 different features are extracted, ranging from time-domain descriptors to frequency-domain ones. The extracted features are classified with three different classifiers (k -NN, SVM, and ANN) using them both alone and stacking their predictions. Experiments have been run on a publicly available database of simulated driving data. The recognition problem is formulated as a binary problem (i.e. stress vs no-stress) in two different setups: five-fold cross validation and leave one subject out. Experimental results showed that the individual signals able to better discriminate among stress and no-stress are BR and PER-EDA, but the best recognition rate is obtained by concatenating the features of all the signals considered. The micro accuracy reached by the best configuration in the first experimental setup is more than 77% (concatenating the features of all the four signals considered and using the stacked classifier), while in the second setup is more than 65%, reflecting the difficulty of the model adaptation to unseen subjects (concatenating the features of all the four signals considered and using the ANN classifier).

As future work we plan to exploit transfer learning techniques to increase the recognition rate in the challenging leave one subject out experimental setup and to investigate the use of 1D Convolutional Neural Networks (CNNs) inspired by famous 2D CNNs [2].

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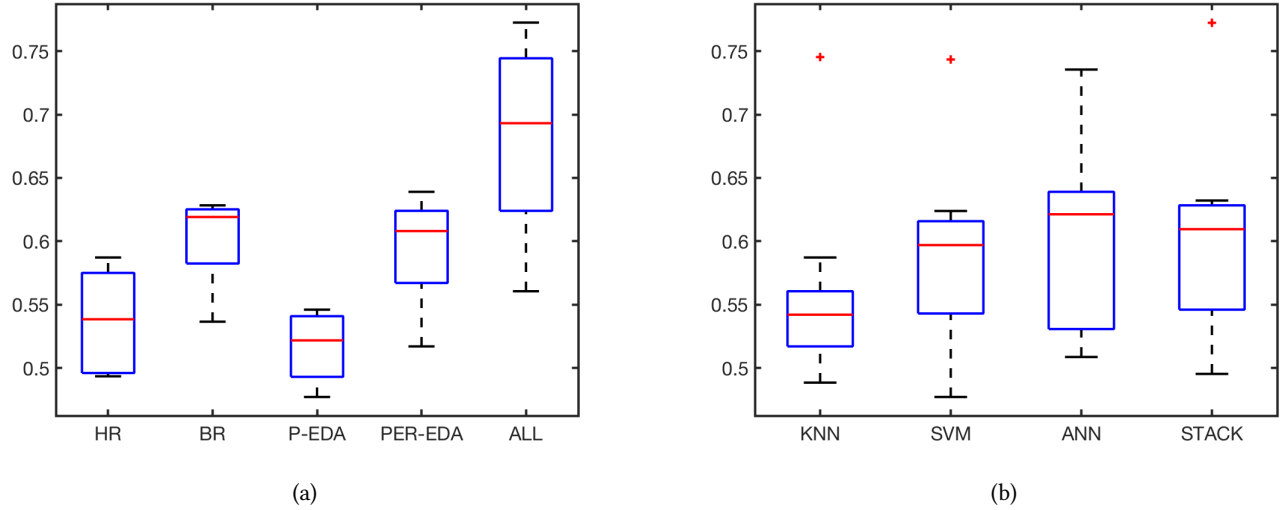
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Table 1: Stress recognition results in terms of micro accuracy ($\pm$ standard deviation) for the 5-fold experimental setup.

	HR	BR	P-EDA	PER-EDA	ALL
KNN	58.72 ($\pm$ 1.59)	55.97 ($\pm$ 1.99)	53.88 ($\pm$ 1.36)	54.54 ($\pm$ 0.77)	74.51 ($\pm$ 1.21)
SVM	56.87 ($\pm$ 1.35)	62.39 ($\pm$ 1.33)	54.31 ($\pm$ 1.36)	61.42 ($\pm$ 1.75)	74.37 ($\pm$ 1.23)
ANN	56.83 ($\pm$ 2.31)	62.12 ($\pm$ 1.14)	53.08 ($\pm$ 1.32)	63.90 ($\pm$ 1.53)	73.55 ($\pm$ 2.11)
STACK	58.14 ($\pm$ 2.17)	62.84 ($\pm$ 1.61)	54.61 ($\pm$ 1.26)	62.67 ($\pm$ 0.90)	77.25 ($\pm$ 1.13)

Table 2: Stress recognition results in terms of micro accuracy ($\pm$ standard deviation) for the one-subject-out experimental setup.

	HR	BR	P-EDA	PER-EDA	ALL
KNN	49.68 ($\pm$ 5.68)	53.66 ($\pm$ 6.17)	48.85 ($\pm$ 5.84)	51.71 ($\pm$ 5.70)	56.06 ($\pm$ 6.96)
SVM	49.35 ($\pm$ 7.44)	60.52 ($\pm$ 9.70)	47.72 ($\pm$ 5.89)	58.88 ($\pm$ 9.11)	61.59 ($\pm$ 7.64)
ANN	50.87 ($\pm$ 4.47)	62.66 ($\pm$ 10.38)	51.29 ($\pm$ 4.20)	62.14 ($\pm$ 9.46)	65.09 ($\pm$ 8.07)
STACK	49.54 ($\pm$ 5.38)	61.71 ($\pm$ 9.29)	49.76 ($\pm$ 5.26)	60.20 ($\pm$ 8.39)	63.22 ($\pm$ 7.52)

**Figure 2: Boxplots of the recognition performance in the two experimental setups together, grouped by the signal used to extract the features (a) and by the classifier used (b).****Table 3: stress vs no-stress recognition accuracy obtained by the best method (stack of classifiers and all physiological signals).**

Folding	stress	no-stress	average
5-fold	82.23 ($\pm$ 1.91)	72.28 ($\pm$ 3.37)	77.25 ($\pm$ 1.13)
SubjOut	74.84 ($\pm$ 25.03)	51.59 ($\pm$ 20.20)	63.22 ($\pm$ 7.52)

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